

Homework #4

Due date: 2026/06/19

遲交則原始成績 1 天*0.9、2 天*0.8、3 天*0.7，>3 天為 0 分

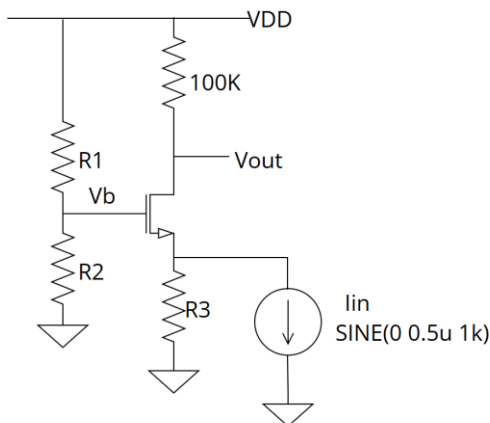
Notice:

- (1)) Please submit your report (*.pdf) and schematic file (*.asc) to the EECLASS system. [-5pt]
- (2) Please name your report as hw3_studentID.pdf [-5pt]
- (3) Please name your schematic file as hw3_studentID.asc. If you submit additional files for specific problems, please name them as hw3_studentID_xxx.asc, where xxx indicates the problem name. For example: hw3_113061000_c.asc and hw3_113061000_d.asc. [-5 pt]
- (4) Please set white background and data trace width = 3 in tutorial. [-5pt]
- (5) You must extract the data of waveform by cursor in tutorial. [-5pt]
- (6) You must attach screenshots of schematic with simulation commands and output waveform in your report. [-5pt]
- (7) List calculation/design process clearly and mark proper information of all plots/figures required.
- (8) Screenshots of the waveforms must include both x-axis and y-axis labels. [-5 pt]

General Specifications:

- Supply Voltage VDD: 1.8 V
- Transistor Model Parameters: $V_{t0} = 0.35 \text{ V}$, $k_p = 200 \mu\text{A/V}^2$
- Performance Targets:
 - Current Gain: $> 0.95 \text{ A/A}$
 - Total Power Consumption: $< 0.6 \text{ mW}$

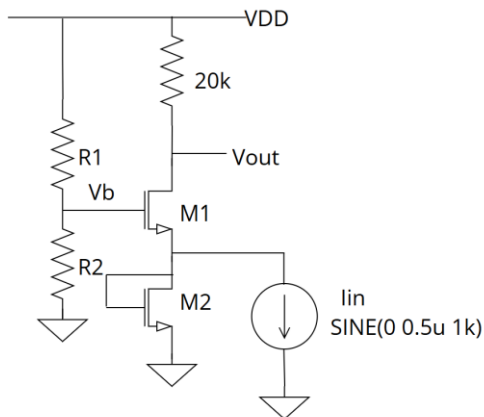
Problem 1:



Consider the Common-Gate architecture shown above. Please follow the steps below to design the current buffer to meet the performance targets.

- Design the main drain current and the biasing current to satisfy the total power consumption requirement. (Note: Please clearly list your calculation steps.)
- Choose the DC node voltages. Given the load resistance = 100 k Ω , please calculate and design R3. Next, using $V_{TO} = 0.35$ V, $K_P = 200$ $\mu\text{A}/\text{V}^2$ design the W/L ratio of M1 to satisfy the current gain requirement. Finally, design R1 and R2 to reach your required bias voltage.
- Draw the schematics of your design in your SPICE simulator and set the component values to those designed in (a)~(b). (Note: Please take a screenshot and submit the schematic file.)
- Simulate the .op to verify your power consumption, and run an AC simulation to plot the input signal current and output load current. Calculate the Current Gain. Does the simulated gain perfectly agree with the > 0.95 A/A requirement? If not, please explain the physical reason and re-design your transistor size or bias current to approach the correct gain! Include the updated screenshots.

Problem 2:



To improve the output impedance and buffering capability, consider the Cascode architecture shown above.

- a.) Similar to Problem 1, design drain current and the biasing current to satisfy the power consumption limit.
- b.) Choose the DC node voltages.. Given the load resistance is now changed to 20 k Ω , please calculate and design the size of the bottom transistor M2. Next, design the W/L ratio of M1 to satisfy the current gain requirement. Finally, design R1 and R2 to reach your required bias. (Note: Please clearly list all your design/calculation steps.)
- c.) Draw the schematics of your Cascode design in your SPICE simulator. (Note: Submit the schematic screenshot and file.)
- d.) Simulate the .op and AC response. Plot the current waveforms and verify the gain. Compare the AC simulation results between Problem 1 and Problem 2. Explain how the Cascode architecture improves the overall current transfer ratio.